

Homework 2

ASEN 5090 Introduction to GNSS

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Problem 1

(a.)

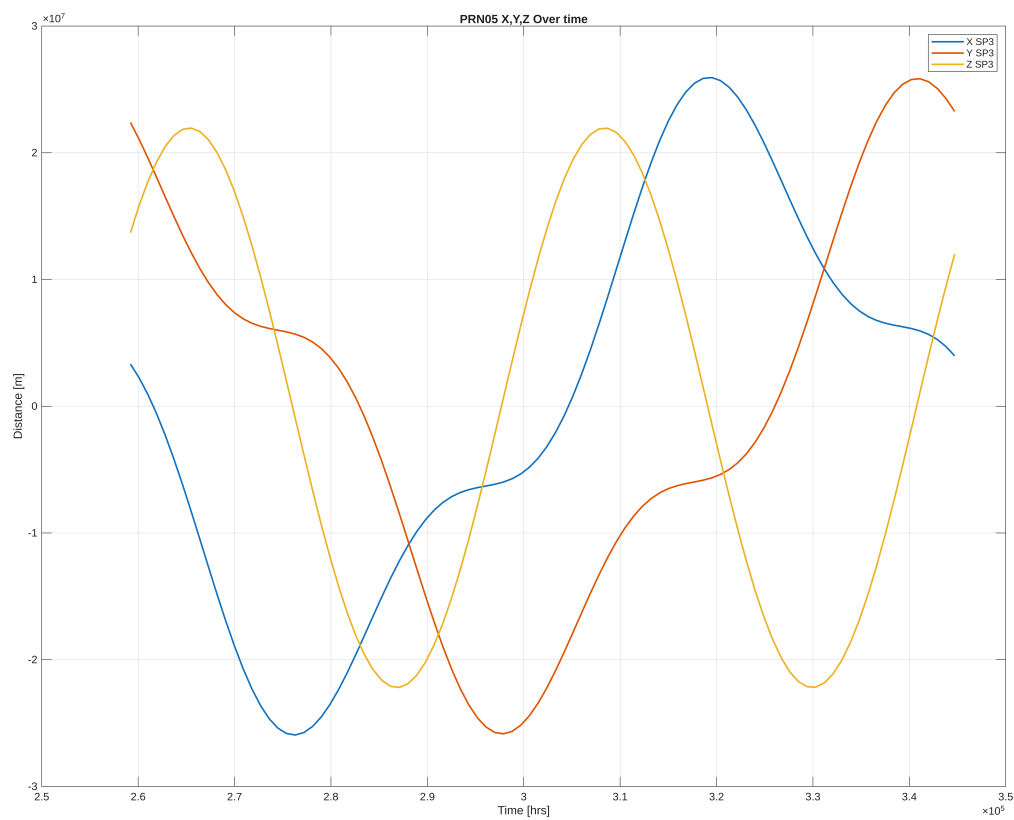


Figure 1: XYZ vs Time of SP3 Output

(b)

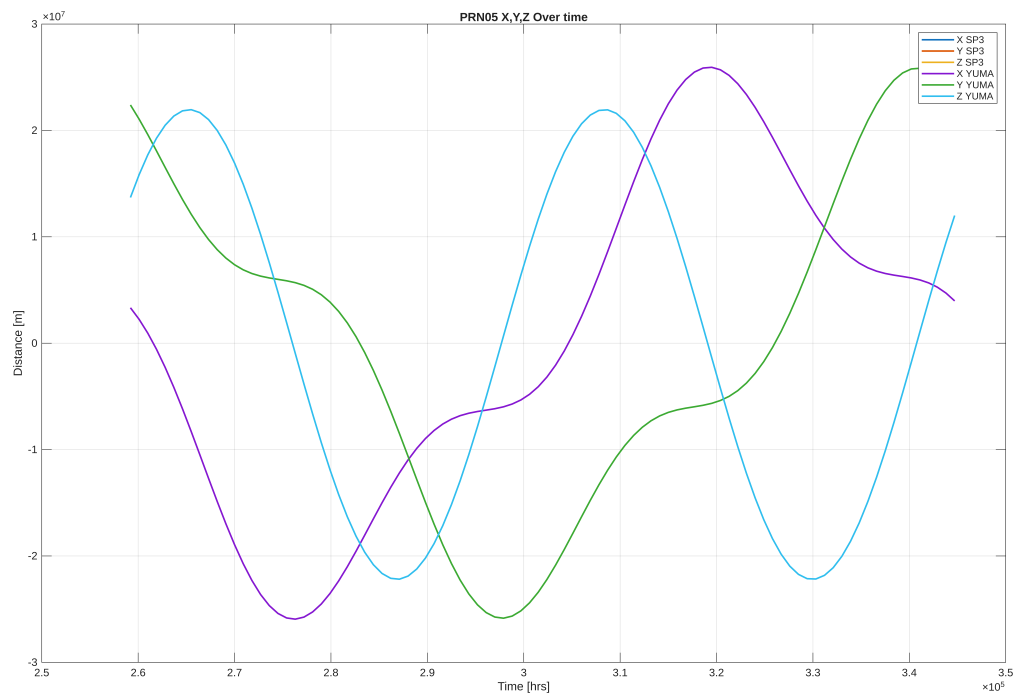


Figure 2: XYZ vs Time of SP3 Output with Almanac

(c.)

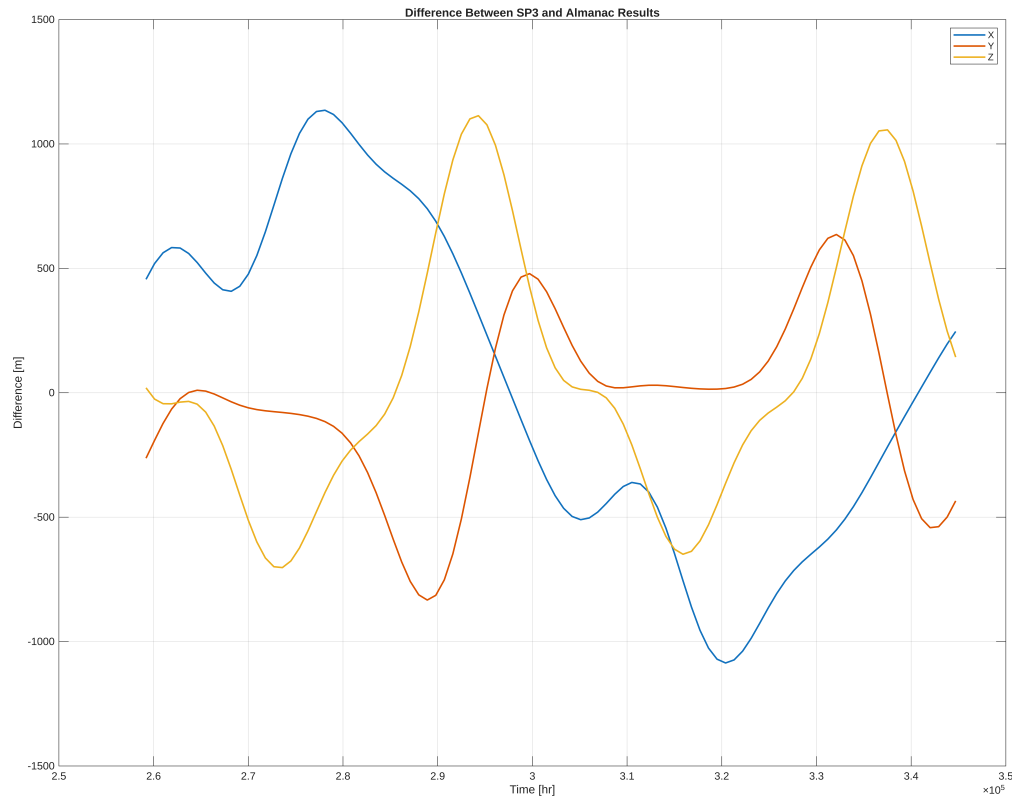


Figure 3: Residuals Between Almanac and SP3

(d.)

When observing the differences between the results of SP3 and the yuma almanac, if we treat SP3 data as our truth data we would have differences of up to 1 km as observed in Figure 3. Although large, if we consider the MEO orbit of the GPS constellations and the fact the observer will be on Earth, the differences in angles which we would use to determine if a satellite is visible would be less than a degree. This distance is not useful for tracking however very useful for checking if satellites are visible or not.

Problem 2

(a.)

Station	Lat (deg)	Lon (deg)	Height (m)	X (m)	Y (m)	Z (m)
NIST	39.995067	-105.262606	1648	-1288398	-4721697	4078625
SMEAD	40.010389	-105.244278	1588	-1286588	-4721009	4079890
EQUATOR	0	-105.262606	-18	-1678998	-6153159	0

Table 1: Station coords

TODO: Explain how we got these LLAs.

(b)

Examine MATLAB function in MATLAB section below. We convert the ECEF vectors of

`r_enu_NIST =`

```
    0
-0.0033
 1.0000
```

`r_enu_EQUATOR =`

```
    0
    0
 1.0000
```

Problem 3

(a.)

(b)

(c.)

(d.)

Problem 4

(a.)

(b)

(c.)